

Survival analysis in infectious disease modelling

From local association in bivariate event time data to the analysis of multivariate serological data

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- ① Prologue
- ② Frailty models to reveal how infectious diseases are transmitted
- ③ Measuring local association between infection times
- ④ Epilogue - other research topics

Prologue

How are infectious diseases transmitted

- ▶ **Infectious disease epidemiology** relies on statistical and mathematical approaches to model the spread of infectious diseases (Anderson and May, 1991)
- ▶ **Transmission process** governed by **(social) mixing patterns** (with contacts deemed relevant for transmission)
- ▶ **Transmission parameters** estimated from **infectious disease data** (serological data, outbreak data, etc.), e.g. force of infection $\lambda(a)$, basic reproduction number R_0 , ...
- ▶ **Force of infection** relates to transmission rates through the so-called “**mass action principle**”; infectious and susceptible individuals are assumed to mix completely in the population

Mass action principle in its simplest form

Homogeneous mixing

Force of infection:

$$\lambda = \beta I$$

- ▶ β : transmission rate
- ▶ I : number of infected individuals

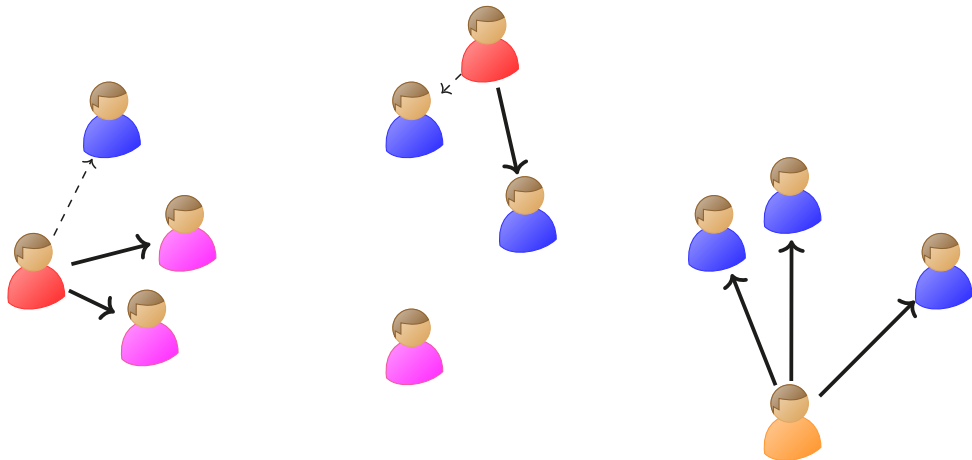
Number of new infections:

$$\lambda S = \beta IS$$

- ▶ S : number of susceptible individuals

Heterogeneity in the acquisition of infections

- Individual variation in susceptibility to infection, infectiousness upon infection (e.g., superspreaders and viral load) and social contact behaviour



- ▶ Variation in **measurable** (e.g., age) and **unmeasurable** (unobserved) (resulting from, e.g., genetic predisposition) factors
- ▶ Impact on **key epidemiological parameters** studied before by, e.g., Coutinho et al. (1999) and Farrington et al. (2001)
- ▶ Recently, investigation of **types of superspreading** on SARS-CoV-2 infections (e.g., Kuylen et al., 2022)
- ▶ Sources of heterogeneity are **not always accounted for** in (statistical or mathematical) infectious disease models
- ▶ **Individual heterogeneity** in acquisition of infections incorporated in mass action principle
- ▶ Use of **survival models** to accommodate (individual) heterogeneity in infectious disease data

Heterogeneous mixing

Force of infection:

$$\lambda(a) = ND \int_0^\infty \beta(a, a') \lambda(a') S(a') \phi(a') da'$$

- ▶ $\beta(a, a')$: effective contact rate
- ▶ $S(a')$: proportion of susceptible individuals of age a'
- ▶ $\phi(a') = L^{-1} \exp\left(-\int_0^{a'} \mu(u) du\right)$
- ▶ L life expectancy, $\mu(a)$ mortality rates, D mean infectious period and N population size

Frailty models to reveal how infectious diseases are transmitted

Assume that for each individual we have

- ▶ **Measured characteristics** $\mathbf{x} = (x_1, \dots, x_p)^T \in \mathbb{R}^{p+1}$
 - ▶ age
 - ▶ sexual behaviour
 - ▶ needle sharing practice
- ▶ **Unobserved (latent) characteristics** $\mathbf{z} = (z_1, \dots, z_q)^T \in \mathbb{R}^{q+1}$
 - ▶ susceptibility
 - ▶ infectiousness
 - ▶ sociability and social environment
 - ▶ behavioural variables (e.g., propensity to use public transport or attend large gatherings)

- ▶ Extended effective contact rates $\beta(\mathbf{x}, \mathbf{z}; \mathbf{x}', \mathbf{z}')$ represent the **per capita rate** at which an infectious individual with characteristics $(\mathbf{x}, \mathbf{z})$ **makes effective contacts** with susceptible individuals with characteristics $(\mathbf{x}', \mathbf{z}')$ (Diekmann and Heesterbeek, 2001)
- ▶ **Multiplicative decomposition** (Farrington et al., 2001) and **separability**:

$$\beta(\mathbf{x}, \mathbf{z}; \mathbf{x}', \mathbf{z}') = \alpha(\mathbf{x}, \mathbf{z}; \mathbf{x}', \mathbf{z}') \beta_0(\mathbf{x}, \mathbf{x}') = w(\mathbf{x}, \mathbf{z}) \beta_0(\mathbf{x}, \mathbf{x}') w'(\mathbf{x}', \mathbf{z}'),$$

where

$$\beta_0(\mathbf{x}, \mathbf{x}') = \int_{\mathbf{z}} \int_{\mathbf{z}'} \beta(\mathbf{x}, \mathbf{z}; \mathbf{x}', \mathbf{z}') f_{\mathbf{Z}}(\mathbf{z}) f_{\mathbf{Z}'}(\mathbf{z}') d\mathbf{z} d\mathbf{z}'$$

is the **average effective contact rate** for individuals with measured characteristics $\mathbf{x}$ and $\mathbf{x}'$

- ▶ Functions $w(\cdot, \cdot)$ and $w'(\cdot, \cdot)$ might differ and, for reasons of identifiability, we assume

$$E_{\mathbf{z}} [w(\mathbf{x}, \mathbf{z})] = E_{\mathbf{z}'} [w'(\mathbf{x}, \mathbf{z}')] = \tilde{w}(\mathbf{x})$$

- ▶ For infections in endemic equilibrium, **force of infection** can be expressed as **integral equation**:

$$\begin{aligned}\lambda(\mathbf{x}, \mathbf{z}) &= \int_{\mathbf{x}'} \int_{\mathbf{z}'} \beta(\mathbf{x}, \mathbf{z}; \mathbf{x}', \mathbf{z}') I(\mathbf{x}', \mathbf{z}') d\mathbf{z}' d\mathbf{x}' \\ &= w(\mathbf{x}, \mathbf{z}) \int_{\mathbf{x}'} \int_{\mathbf{z}'} \beta_0(\mathbf{x}, \mathbf{x}') w'(\mathbf{x}', \mathbf{z}') I(\mathbf{x}', \mathbf{z}') d\mathbf{z}' d\mathbf{x}' =: w(\mathbf{x}, \mathbf{z}) \lambda_0(\mathbf{x}),\end{aligned}$$

with $\lambda_0(\mathbf{x})$ the so-called **baseline force of infection**

- ▶ This defines a **frailty model** for the equilibrium force of infection
- ▶ Different **sources of unobserved heterogeneity**, expressed in terms of $w(\mathbf{x}, \mathbf{z})$ and $w'(\mathbf{x}', \mathbf{z}')$, and their impact on, e.g., R_0 can be studied
- ▶ **Proportional hazards model**: $w(\mathbf{x}, \mathbf{z}) = \mathbf{z} \tilde{w}(\mathbf{x}) = \mathbf{z} \exp(\boldsymbol{\beta}^T \mathbf{x})$

Connection with (simple) mathematical models

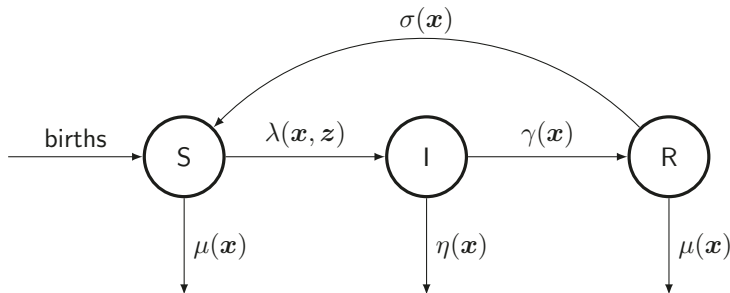


Figure: Individuals are born into the susceptible class **S** and move to the infected state **I** at rate $\lambda(\mathbf{x}, z)$, after which they recover and move to **R** at rate $\gamma(\mathbf{x})$. Subsequently, individuals lose protective immunity and move back to **S** at replenishment rate $\sigma(\mathbf{x})$. All individuals are subject to natural mortality at rate $\mu(\mathbf{x})$ and infected individuals to an additional disease-related mortality at rate $\alpha(\mathbf{x})$ ($\eta(\mathbf{x}) = \mu(\mathbf{x}) + \alpha(\mathbf{x})$).

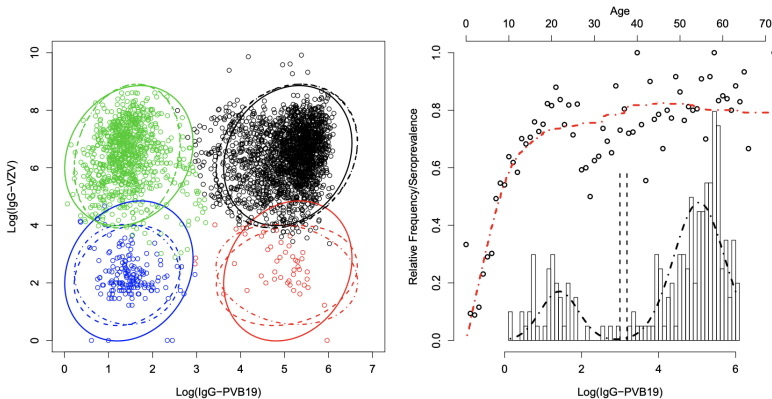
- ▶ Bivariate serological data $(y_1, y_2, \mathbf{x})$ under endemic equilibrium with y_i ($i = 1, 2$) representing the immunological status at sampling time
- ▶ Solving the system of (ordinary) differential equations (under endemic equilibrium) provides a (numerical) solution for $I(\mathbf{x}', \mathbf{z}')$
- ▶ Marginalisation of $I(\mathbf{x}', \mathbf{z}')$ (or equivalently $S(\mathbf{x}', \mathbf{z}')$) in terms of frailty distribution (characterised by $f_{\mathbf{Z}}(\mathbf{z})$) leads to (tractable) expressions used in ML inference
- ▶ Extension to bivariate setting with infection-specific hazard functions ($i = 1, 2$):

$$\lambda_i(\mathbf{x}, \mathbf{z}_i) = w_i(\mathbf{x}, \mathbf{z}_i) \lambda_{i0}(\mathbf{x})$$

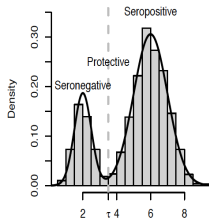
- ▶ Different types of frailty models induced by choice of bivariate frailty distribution $f_{\mathbf{Z}_1, \mathbf{Z}_2}$
 - ▶ Shared frailty model structure: $\mathbf{Z}_1 = \mathbf{Z}_2 = \mathbf{Z}$ (heterogeneity in contact rates)
 - ▶ Correlated frailty model structure: $\mathbf{Z}_1, \mathbf{Z}_2$ (heterogeneity in susceptibility & infectiousness)

- ▶ Non-immunising infections (Abrams and Hens, 2015): SIRS vs. SIR model structures
- ▶ Overdispersed frailty models (Abrams et al., 2017): partial marginalisation over additional stratum-specific random effects
- ▶ Time-varying individual heterogeneity (Abrams et al., 2018): $z_i(t)$
- ▶ Correlated gamma frailty models (Martins et al., 2019): more complex bivariate frailty distributions f_{Z_1, Z_2} instead of using variable-in-common method (Yashin et al., 1995)
- ▶ General frailty model for multivariate serological data (Tran et al., 2020)

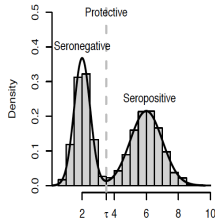
- Mixture approach for continuous antibody titer data elucidating waning of humoral immunity



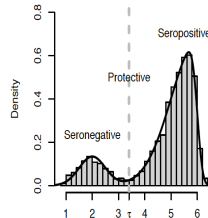
Recent and ongoing research



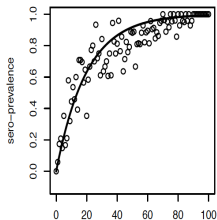
Log-transformed IgG antibodies



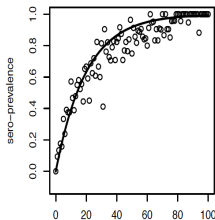
Log-transformed IgG antibodies



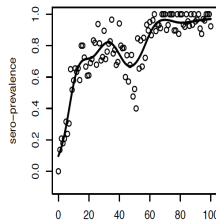
Log-transformed IgG antibodies



Age (in years)



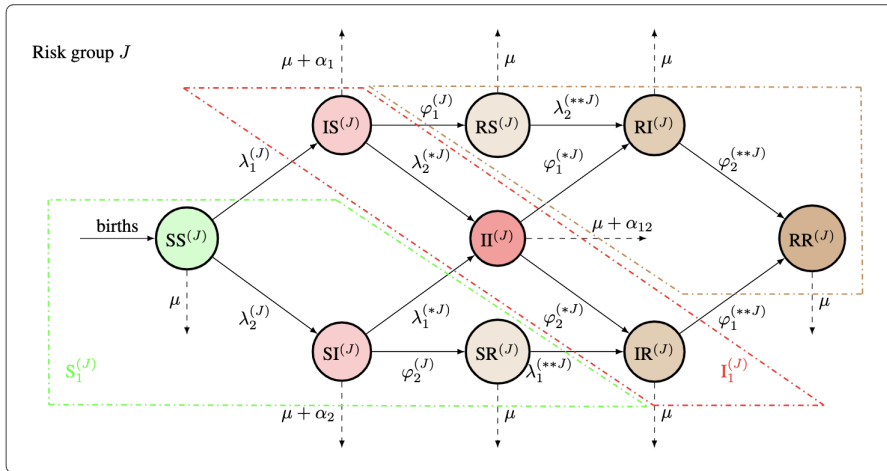
Age (in years)



Age (in years)

- ▶ Mixture approach for **continuous antibody titer data** elucidating **waning of humoral immunity** (Martins et al., 2025)
- ▶ How do **statistical frailty approaches** align with **mathematical models** in the presence of individual heterogeneity? Do we measure what we want to measure? $\Rightarrow$ Discrete versus continuous frailty models (Grosso et al., *in preparation*)

Towards more complex models



Measuring local association between infection times

- ▶ In infectious disease epidemiology, when unraveling transmission routes, one is often interested in studying **association between infection times T_1 and T_2**
- ▶ Strength of association can be expressed in terms of **global and local association measures**
- ▶ **Cross-ratio function** was introduced by Clayton (1978):

$$\begin{aligned}\theta(t_1, t_2) &= \frac{\lambda_1(t_1|T_2 = t_2)}{\lambda_1(t_1|T_2 > t_2)} = \frac{\lambda_2(t_2|T_1 = t_1)}{\lambda_2(t_2|T_1 > t_1)} \\ &= \frac{S(t_1, t_2)\partial_{12}^2 S(t_1, t_2)}{\partial_1^2 S(t_1, t_2)\partial_2^2 S(t_1, t_2)}\end{aligned}$$

- ▶ **Interpretation** (Oakes, 1982, 1986):
 - ▶ $\theta(t_1, t_2) = 1$: infection times T_1 and T_2 are **independent**
 - ▶ $\theta(t_1, t_2) > 1$: infection times T_1 and T_2 are **positively associated**
 - ▶ $\theta(t_1, t_2) < 1$: infection times T_1 and T_2 are **negatively associated**

- **Conditional frailty distribution**, i.e., the distribution of Z conditional on no infection at least up to time t , is characterised by the density function

$$f_Z(z | T > t) = \frac{S(t | z) f_Z(z)}{S(t)},$$

where f_Z is the (unconditional) probability density function of individual frailty Z

- **Relative frailty level**, defined as

$$Z_t = \frac{Z}{E(Z | T > t)},$$

gives rise to the so-called **relative frailty distribution** in survivors up to time t

- ▶ Relative frailty variance is equal to

$$\text{RFV}(t) = \text{Var}(Z_t \mid T > t) = \frac{\text{Var}(Z \mid T > t)}{E(Z \mid T > t)^2}$$

- ▶ Different choices of the frailty distribution f_Z lead to different expressions of the $\text{RFV}(t)$
- ▶ Extension for two infection times T_1 and T_2 and shared frailty level Z :

$$\text{RFV}(t_1, t_2) = \frac{\text{Var}(Z \mid T_1 > t_1, T_2 > t_2)}{E(Z \mid T_1 > t_1, T_2 > t_2)^2}$$

- ▶ One can show that (Farrington et al., 2012):

$$\theta(t_1, t_2) = \text{RFV}(t_1, t_2) + 1$$

- ▶ Generalisation for correlated frailty models possible in terms of $\text{RFCV}(t_1, t_2)$

- ▶ Cross-ratio function can be considered for **diagnostics regarding frailty distribution** (or copula function) used to describe dependence in bivariate survival data
- ▶ **Some earlier work:**
 - ▶ Hsu and Prentice (1996): crude nonparametric estimator for the cross ratio function
 - ▶ Nan et al. (2006): parametric piecewise constant estimation in rectangular regions
 - ▶ Li and Lin (2006) & Li et al. (2008): copula-based approaches
 - ▶ Glidden (2007): implicit nonparametric estimator for the cross ratio function, though very rough behaviour
 - ▶ Hu et al. (2011): parametric polynomial models for the (log-transformed) cross ratio function applied to right-censored bivariate survival data
- ▶ **Our work:**
 - ▶ Smooth Bernstein-based estimator of conditional hazards in $\theta(t_1, t_2)$ (Abrams et al., 2020)
 - ▶ Direct Bernstein-based estimation of survival function and its derivatives in $\theta(t_1, t_2)$ (Abrams et al., 2024)

Bernstein-based estimator of $\theta(t_1, t_2)$

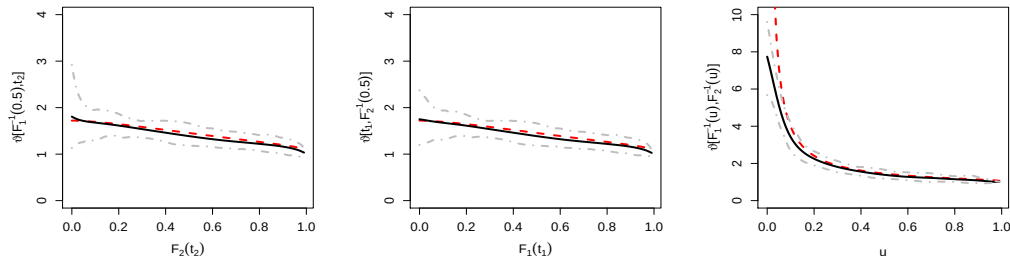


Figure: Gumbel copula example (Abrams et al., 2020)

- ▶ Bernstein-based estimators for $\theta(t_1, t_2)$ under different types of (right-)censoring
- ▶ Semi-parametric estimators based on P- and O-splines accounting for boundary effects
- ▶ Goodness-of-fit measures based on $\theta(t_1, t_2)$ to assess the choice of (bivariate) frailty distribution
- ▶ Serological data: estimation of $\theta(t_1, t_2)$ under Type I interval-censored (current status) data and inferring time of infection for serological data
- ▶ Related work on
 - ▶ Quantiles of the conditional residual lifetime distribution (Abrams et al., 2021)
 - ▶ Estimation of risk ratios for bivariate survival data (Abrams et al., 2022)
 - ▶ Nonparametric estimators for the quantiles of the conditional residual lifetime distribution (Abrams et al., 2025, *in revision*)

Some recent work illustrated (*in preparation*)

- ▶ Bernstein-based estimators for $\theta(t_1, t_2)$ under different types of (right-)censoring
- ▶ **Semi-parametric estimators based on P- and O-splines** accounting for boundary effects (**joint work with Miguel Beynaerts, Niel Hens and Anneleen Verhasselt**)
- ▶ **Goodness-of-fit measures** based on $\theta(t_1, t_2)$ to assess the choice of (bivariate) frailty distribution
- ▶ Serological data: estimation of $\theta(t_1, t_2)$ under **Type I interval-censored (current status) data** and **inferring time of infection for serological data**
- ▶ Related work on
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- ▶ Consider a sample $(X_{1i}, \Delta_{1i}, X_{2i}, \Delta_{2i})$, $i = 1, \dots, n$, of bivariate time-to-event data from $(X_1, \Delta_1, X_2, \Delta_2)$ with for $j = 1, 2$:
 - ▶ $X_j = \min(T_j, C_j)$ **observation times**,
 - ▶ $\Delta_j = I(T_j \leq C_j)$ **censoring indicators**, and
 - ▶ (C_1, C_2) **bivariate censoring times**,under **independent (and non-informative) right censoring**
- ▶ Hu et al. (2011) proposed a **pseudo-partial likelihood** approach for the estimation of the CRF by considering T_1 as a covariate in a Cox PH model
- ▶ **Ratio of hazards** of event two in groups $\{i \mid T_{1i} = t_1\}$ and $\{i \mid T_{1i} > t_1\}$ equals the CRF

- ▶ CRF is **symmetric** with respect to switching the roles of T_1 and T_2 : symmetrization step in **log-likelihood** to guarantee symmetry for the estimator:

$$l(\beta) = \sum_i \left[l_i^{(1)}(\beta) + l_i^{(2)}(\beta) \right], \quad (1)$$

where β are parameters associated with the $\log \theta(t_1, t_2)$

- ▶ Pseudo-partial log-likelihood function (1) is maximized to obtain the Maximum Likelihood (ML) estimates $\hat{\beta} = \operatorname{argmax}_{\beta} l(\beta)$

- ▶ **Approach of Hu et al. (2011):**

$$\log \theta(t_1, t_2) = \sum_{r+s \leq 3} \beta_{rs} t_1^r t_2^s$$

- ▶ **Our approach**

$$\log \theta(t_1, t_2) = \sum_{r=1}^{r'} \sum_{s=1}^{s'} \beta_{rs} B_1^r(t_1) B_2^s(t_2)$$

Flexible estimation of $\log \theta(t_1, t_2)$ requires a large parameter space Identifiability might not be guaranteed. We introduce **penalization** into the (log-)likelihood

$$l_{\lambda}(\beta) = l(\beta) - \frac{\lambda_1}{2} \beta^T S_1 \beta - \frac{\lambda_2}{2} \beta^T S_2 \beta$$

with

- ▶ $\lambda = (\lambda_1, \lambda_2)$ a vector of smoothing parameters to be chosen in a data-driven way, and
- ▶ S_i ($i = 1, 2$) the corresponding positive semidefinite penalty matrices

Possible penalties:

- ▶ **Discrete (d -order difference) penalties:** P-splines (Eilers & Marx, 1996)
- ▶ **Integrated squared derivative penalties:** O-splines (O'Sullivan, 1986)

General idea from Wahba (1983):

$$\beta|\lambda \sim N\left(\mathbf{0}, \left(\sum_k \lambda_k \mathbf{S}_k\right)^{-1}\right).$$

This leads to the Laplace approximation of the (penalized) log-likelihood

$$\mathcal{V}(\lambda) = l(\beta) - \frac{1}{2}\beta^T S_\lambda \beta + \frac{1}{2} \log |S_\lambda|^+ - \frac{1}{2} \log |\mathcal{H}_\lambda| + \frac{\mathcal{M}_p}{2} \log(2\pi),$$

where $S_\lambda = \sum_i \lambda_i \mathbf{S}_i$, $|\cdot|^+$ the product of all positive eigenvalues, $\mathcal{H}_\lambda$ the negative Hessian of the penalized log-likelihood and $\mathcal{M}_p$ the number of zero eigenvalues of S_λ when all λ_i are strictly positive

Wood and Fasiolo (2017) proposed a **simple iterative procedure** to maximize the **Laplace approximated (penalized pseudo-partial) log-likelihood function**

$$\lambda_i^* = \frac{\text{tr}(S_{\lambda}^{-1} S_i) - \text{tr}(\mathcal{H}_{\lambda}^{-1} S_i)}{\hat{\beta}^T S_{\lambda} \hat{\beta}} \lambda_i.$$

Procedure:

- 1 Choose initial $\lambda_{(0)}$
- 2 Maximize penalized likelihood to obtain $\hat{\beta}|\lambda_{(0)}$
- 3 Update $\lambda_{(0)}$ to $\lambda_{(1)}$ using $\hat{\beta}|\lambda_{(0)}$
- 4 Re-estimate $\hat{\beta}|\lambda_{(1)}$
- 5 Check convergence criteria
- 6 Repeat steps 2–5 until convergence

We fit P-splines with dimension 17×17 to samples of size 500, 750 and 1000

Setting 1:

- ▶ Dependence through **Frank copula** with association parameter α : $(u_1, u_2) \sim \mathcal{C}_\alpha$
- ▶ Margin $F_T \sim \text{Unif}(0, 5)$ such that $(T_1, T_2) = (F_T^{-1}(u_1), F_T^{-1}(u_2))$
- ▶ Independent bivariate censoring times $C_j \sim \text{Exp}(\lambda)$ ($j = 1, 2$)

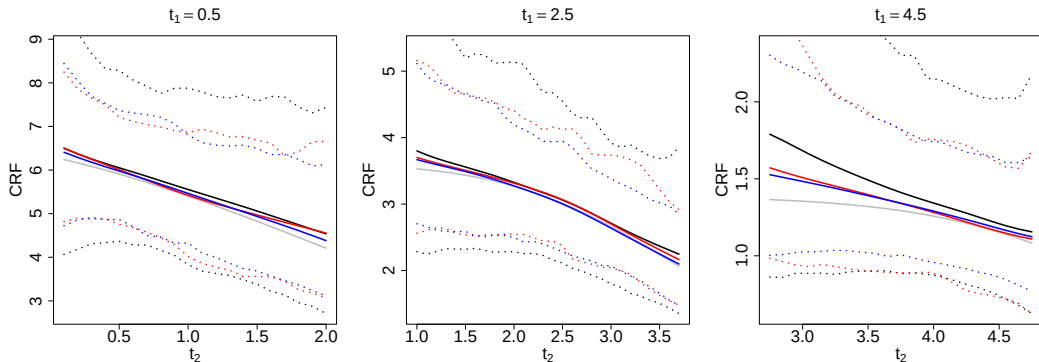
X_1/X_2		50%		
	Censoring	n	$\log \theta$	$\log \hat{\theta}$ (SE_e , SE_m , CP)
50%	20%	500	1.103	1.100 (0.193, 0.133, 94.6%)
		750		1.108 (0.159, 0.109, 95.0%)
		1000		1.090 (0.132, 0.095, 93.8%)
	40%	500		1.109 (0.292, 0.184, 94.8%)
		750		1.093 (0.206, 0.147, 95.2%)
		1000		1.087 (0.178, 0.127, 95.0%)

- ▶ Estimator performs well in terms of bias and CP in regions with sufficient uncensored observations (example at the main diagonal)
- ▶ Model-based standard error estimates underestimate empirical standard errors

X_1/X_2		75%		
	Censoring	n	$\log \theta$	$\log \hat{\theta}$ (SE_e , SE_m , CP)
25%	20%	500	0.743	0.991 (0.443, 0.279, 91.2%)
		750		0.944 (0.289, 0.216, 91.0%)
		1000		0.853 (0.252, 0.187, 93.8%)
	40%	500		1.184 (0.751, 0.422, 92.4%)
		750		0.987 (0.494, 0.313, 92.4%)
		1000		0.938 (0.390, 0.256, 93.0%)

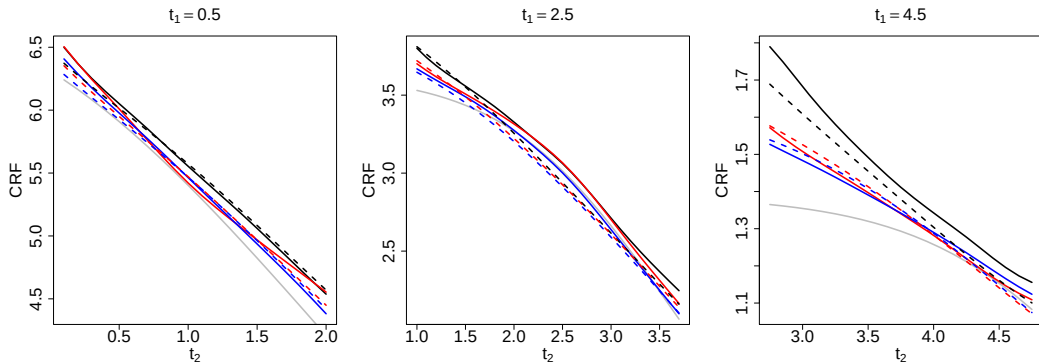
- Boundary issues and few uncensored observations in boundary area reduce finite-sample performance (compared to more parsimonious parametric alternatives)

Selected results: Frank copula with P-splines



P-spline fit with 20% censoring for $n = 500, 750, 1000$

Selected results: Frank copula with P-splines



P-spline model compared to parametric model from Hu et al. (2011) with 20% censoring for
 $n = 500, 750, 1000$

Caribbean, Central and South America Network for HIV research (CCASAnet)

6691 HIV-positive patients in 5 different countries are treated with antiretroviral therapy (Eden et al., 2023)

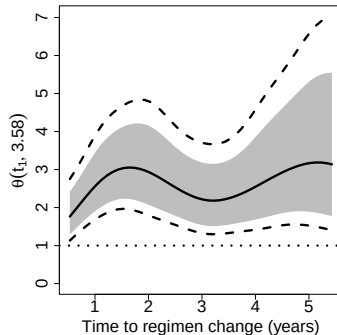
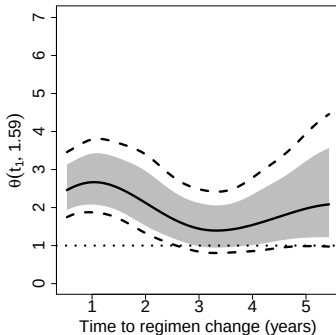
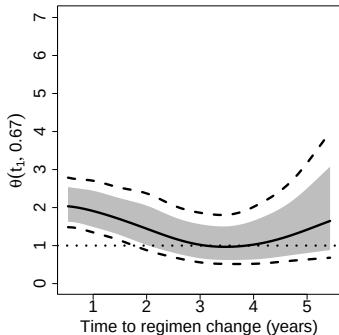
Two event times of interest (under bivariate right censoring):

- ▶ t_1 : time to regimen change (years)
- ▶ t_2 : time to viral failure (years)

Goal: We investigate the time-varying dependence between the two event times in countries A and B using our novel estimator with bivariate P-splines of dimension 17×17 .

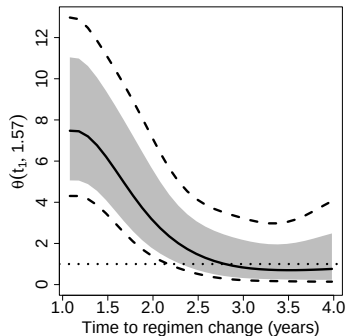
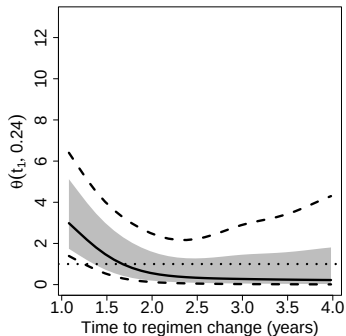
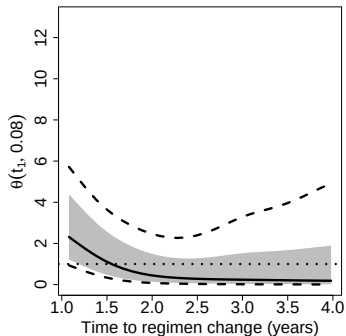
Real-life data example: CCASAnet

Country A



Real-life data example: CCASAnet

Country B



- ▶ Quantification of the uncertainty around point estimates of the CRF:
 - ▶ **Symmetrization** implies model-based standard errors underestimating the empirical standard error estimates
 - ▶ Correction of the model-based standard error using **sandwich estimators**
 - ▶ **Alternative:** bootstrap approach albeit computationally expensive
 - ▶ Uncertainty related to the **estimation of the optimal smoothing parameters** (can directly be accounted for in a bootstrap approach)
- ▶ **Lack-of-fit measures** in context of conditional modeling approaches
- ▶ **Model extensions:**
 - ▶ Inclusion of **covariates** (conditional CRF estimation)
 - ▶ Use of CRF as measure for surrogate marker validation in **(semi-)competing risk settings**

Epilogue - other research topics

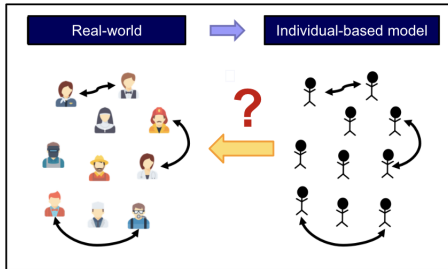
- ▶ Follow-up study in which participants are asked to come to the clinic to get examined for a specific event (infection) during **scheduled routine visits**
- ▶ **Unscheduled clinical visits** can occur when participants develop symptoms for the event (disease) under consideration
- ▶ If the event of interest triggers outcome assessment in between pre-scheduled follow-up visits, the **outcome and observation-time process are dependent**, often referred to as **outcome-dependent sampling** (ODS, see Tan, 2014)
- ▶ Conventional longitudinal methods to analyse repeated measurement over time assume independence of both processes $\Rightarrow$ **not appropriately accounting for ODS** leads to **biased estimation of (epidemiological) quantities** of interest
- ▶ **Malaria application**: Mugenyi et al. (2017), Martins et al. (2022), Martins et al. (2025).

- ▶ **Serial serological survey data** collected at different cross-sectional time points: estimation of age-time specific force of infection (Nagelkerke, 1999; Beynaerts et al., *in preparation*)
- ▶ Extension to encompass **(time-varying) individual heterogeneity** in acquisition of infections
- ▶ **Sero-surveillance** data used to assess and **quantify outbreak risk** (Abrams et al., 2014, Hens et al., 2015): new approaches in the context of measles (in collaboration with *Department Zorg, Flanders*)

- ▶ **Within-host modeling** of the effect of Ebola (booster) vaccination on humoral immunity among HCWs in the DRC (Ebovac trial) (Garcia-Fogeda et al., 2025)
- ▶ **Identification of CoPs** in the context of SARS-CoV-2 (Debie et al., 2025; Garcia-Fogeda et al., *submitted*): joint modeling of time to BTI and humoral immune response
- ▶ Survival analysis of **repeated measurement of time to sputum culture positivity** in the context of multi-drug resistant TB (Van Rie et al., 2024; De Vos et al., 2025; Van Rie et al., *under review*)

Calibration of individual-based models

- ▶ ACCELERATE project (<https://www.accelerate-epi.be/>)
- ▶ Translational framework combining health economics, model inference and reinforcement learning for individual-based models of disease spread



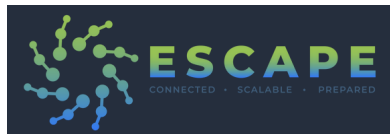
Background image adapted from:
<https://www.lib.ncsu.edu/workshops/introduction-agents-based-modelling-talk>

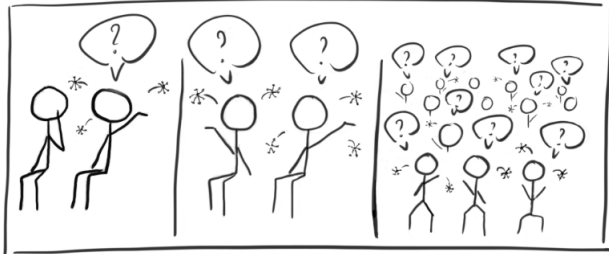


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